Understanding PublicState

A gentle, jargon-free guide to the numbers every trader in the simulator can see

The order_book project

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Abstract

In the simulator, every participant (noise traders, speculators, market makers, and our learning agent) reads the *same* summary of the market at each time step. That summary is the PublicState object. This guide explains, from scratch, what each variable in it means, how it is computed, and why the simulator bothers to track it. No finance background is needed; if you can read a short formula, you can follow everything here.

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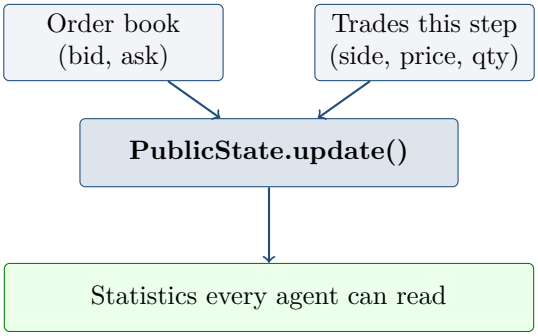
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1 The big picture

A *limit order book* is a list of offers: buyers say "I will pay at most p", sellers say "I will sell for at least p". The highest buy offer is the **best bid**; the lowest sell offer is the **best ask**. A *trade* happens when someone is willing to accept the other side's price right away. Such a trader is called **aggressive** because it takes liquidity instead of waiting.

At every simulation step, after all orders are processed, `PublicState.update(...)` is called exactly once. It looks at the book and at what traded, and refreshes a handful of statistics. Because it is computed once and shared, all agents react to the same information, and nobody sees anything private such as *who* traded.



The variables fall into six families:

Family	Question it answers
Price and book	Where is the market right now, and how expensive is it to trade?
Moving averages	What is the "typical" recent price level?
Volatility	How jumpy is the price?
Volume	How much is being traded?
Momentum	Is the price trending up or down?
Order-flow balance	Are buyers or sellers pushing harder?
Attention	Is the market unusually active right now?

2 A key tool: the exponential moving average

Most variables below use an **exponentially weighted moving average** (EWMA, or EMA). It is a running average that gives more weight to recent data and slowly forgets the past. The update is a single line:

$$\text{avg}_{\text{new}} \;=\; \text{avg}_{\text{old}} + \alpha\,(x_{\text{new}} - \text{avg}_{\text{old}}), \qquad \alpha = \frac{2}{\text{span}+1}. \tag{1}$$

In plain words

Think of the average as a thermometer reading that moves a fraction α of the way toward each new observation. A *short span* gives a big α: the average reacts quickly. A *long span* gives a small α: the average is smooth and slow.

Tiny example

Span $= 9$ gives $\alpha = 0.2$. If the average is 100 and the new value is 110, the average moves to $100 + 0.2 \cdot 10 = 102$.

Why it is used

It needs only *one stored number* per statistic (no history list), so each update costs the same tiny amount of work however long the simulation runs.

The simulator uses two spans, set in `PublicStateConfig`: a **fast** one (default 10 steps) and a **slow** one (default 100 steps). Comparing the two tells us whether *now* looks different from *normal*. This idea comes back again and again below.

3 Price and book variables

3.1 `best_bid`, `best_ask`, `spread`

The best bid and best ask are read directly from the book. The spread is their gap:

$$\text{spread} = \text{best_ask} - \text{best_bid}.$$

Why it is used

The spread is the immediate cost of trading: buying at the ask and selling at the bid loses the spread. Market makers use it to decide where to quote, and execution strategies use it to judge how expensive it is to be impatient.

3.2 `mid`

The **mid-price** is the midpoint of the best quotes:

$$\text{mid} = \frac{\text{best_bid} + \text{best_ask}}{2}.$$

If one side of the book is empty, it falls back to the `last` trade price, and if there has been no trade, it keeps the previous mid.

Why it is used

There is no single "true" price, but the mid is a smooth, fair-looking reference that does not jump around every time a trade bounces between bid and ask. Every statistic below is built on it.

3.3 `last`

The price of the most recent trade (or `None` if nothing has traded yet). It is mostly used as the fallback for the mid described above.

4 Moving averages of price

4.1 ma_fast and ma_slow

Two EWMAs (Eq. 1) of the mid-price, one with the fast span and one with the slow span.

In plain words

`ma_fast` hugs the current price. `ma_slow` is a lazy long-term anchor. When the two differ, the price has recently moved away from its long-run level.

Why it is used

They give agents a cheap estimate of "where the price usually is". For example, a mean-reverting speculator may buy when the price is well below the slow average.

5 Volatility

5.1 var_fast and var_slow

Volatility measures how much the price jumps around. At each step we compute the mid-price change $\Delta = \text{mid}_t - \text{mid}_{t-1}$ and take an EWMA of its square:

$$\text{var} \leftarrow \text{var} + \alpha(\Delta^2 - \text{var}).$$

The unit is *ticks*^{2} *per step* (a tick is the smallest allowed price increment). Both start at 1 to avoid dividing by zero later.

Tiny example

If the price moves $+2, -2, +2, -2$ ticks, $\Delta^2 = 4$ every time, so the variance tends to 4. A quiet market with moves of 0 or ± 1 has variance below 1.

Why it is used

Volatility is the market's mood: wide quotes and cautious behaviour in turbulent times, tight quotes in calm times. The *ratio* of the fast to the slow value flags sudden regime changes (Section 9).

6 Volume

6.1 step_volume

The total aggressive volume traded in the current step: buy_volume + sell_volume.

6.2 volume_fast and volume_slow

EWMAs of `step_volume` at the two horizons.

Why it is used

Volume shows how much participation there is. A burst of volume with `volume_fast` far above `volume_slow` means something is happening, and that is the second ingredient of the attention multiplier.

7 Momentum

7.1 momentum

`momentum` is a *list*, one entry for each (fast, slow) pair in `lookbacks`. The defaults are $(5, 30)$, $(10, 60)$ and $(20, 120)$. For each pair we keep two EWMAs of the mid-price with those spans, and define

$$\text{momentum}_k = \text{EMA}_{\text{fast},k}(\text{mid}) - \text{EMA}_{\text{slow},k}(\text{mid}).$$

In plain words

If the price has been rising, the quick average sits above the slow one, so momentum is **positive**. If it has been falling, momentum is **negative**. A value near zero means no clear trend. Using three pairs gives a short-, medium- and long-term view of the trend.

Why it is used

Trend-following traders buy when momentum is positive and sell when it is negative. Because many agents react to it, it also makes the simulated market behave more like real ones, where trends can feed on themselves.

8 Signed aggressive flow

8.1 buy_flow, sell_flow

Rolling totals of aggressive buy volume and aggressive sell volume over the last `flow_window` steps (default 20). Internally, a circular buffer stores the last 20 values; when a new step arrives it overwrites the oldest one and the running totals are adjusted by (new − old). That makes each update $O(1)$.

8.2 imbalance

$$\text{imbalance} = \frac{\text{buy_flow} - \text{sell_flow}}{\text{buy_flow} + \text{sell_flow}} \in [-1, 1],$$

and 0 if nothing has traded.

Tiny example

imbalance $= +1$: only buyers were aggressive. 0: perfectly balanced. -0.5: sellers did three times as much aggressive volume as buyers.

Why it is used

Random trading is roughly balanced. A big institution splitting a huge order into many small ones (a *metaorder*) pushes the flow persistently one way. This variable is the raw footprint that detectors and liquidity providers look at to spot such activity, which is exactly what the research project studies. The class itself only *computes* the numbers; interpreting them is left to `order_book.detectors`.

8.3 step_trades

The list of anonymous aggressive trades of the current step, each as `(side, price, quantity)`. Trader identities are deliberately not included.

Why it is used

It gives agents and detectors trade-by-trade detail while preserving anonymity, as on a real exchange tape.

9 The attention multiplier

9.1 `attention`

This one combines volatility and volume into a single activity dial. First compare "now" with "normal":

$$r_{\text{vol}} = \sqrt{\frac{\text{var_fast}}{\text{var_slow}}}, \qquad r_{\text{volume}} = \frac{\text{volume_fast}}{\text{volume_slow}}.$$

Each ratio is 1 when things are normal, above 1 when the market is busier than usual, and below 1 when it is quieter. Then

$$\text{attention} = \text{clamp}\Big(1 + \tfrac{1}{2}(r_{\text{vol}} - 1) + \tfrac{1}{2}(r_{\text{volume}} - 1),\ 0.5,\ 3\Big).$$

Tiny example

Volatility is 50% above normal ($r_{\text{vol}} = 1.5$) and volume is double ($r_{\text{volume}} = 2$): $1+0.25+0.5 = 1.75$. Agents are $1.75\times$ as active as usual. A very quiet market bottoms out at 0.5, and a frenzy is capped at 3.

Why it is used

Real markets show **activity clustering**: busy periods attract more activity, and calm periods stay calm. Agents multiply their order-arrival rates by `attention`, so the simulator reproduces that behaviour. The clamp keeps it from exploding or dying out.

9.2 `warmed_up`

The slow averages need time to become meaningful. Early on they are still close to their initial values, so the ratios above would be nonsense. `warmed_up` turns `True` once $n \cdot \alpha_{\text{slow}} > 1$, i.e. after roughly one slow span of data. Until then, `attention` stays at 1.

10 Bookkeeping

`step_count`	The number of updates so far.

11 Cheat sheet

Variable	What it is	Typically used for
`best_bid / best_ask`	Best buy and sell offers	Quoting, trading cost
`spread`	Ask minus bid	Cost of trading now
`mid`	Midpoint of the quotes	Reference price
`last`	Last trade price	Fallback price
`ma_fast / ma_slow`	Smoothed price, short and long	Mean-reversion signals
`var_fast / var_slow`	Smoothed squared price change	Volatility regime
`step_volume`	Volume traded this step	Activity measure
`volume_fast / slow`	Smoothed volume	Activity regime
`momentum[k]`	Fast EMA − slow EMA, per pair	Trend following
`buy_flow / sell_flow`	Rolling aggressive volume per side	Pressure measurement
`imbalance`	Normalised buy/sell difference in $[-1, 1]$	Spotting one-sided execution
`step_trades`	Anonymous trades this step	Tape-level detail
`attention`	Activity multiplier in $[0.5, 3]$	Activity clustering
`warmed_up`	Slow averages are reliable	Guarding ratios
`step_count`	Number of updates	Bookkeeping

12 Default parameters

All parameters live in `PublicStateConfig` (`order_book/config.py`).

Parameter	Default	Meaning
`fast`	10	EWMA span of the fast price, variance and volume statistics
`slow`	100	EWMA span of the slow statistics
`lookbacks`	(5,30), (10,60), (20,120)	(fast, slow) span pairs for momentum
`flow_window`	20	Steps kept in the rolling signed-flow window